```
services:
  some-service:
    deploy:
      resources:
        limits:
          cpus: "0.1"
          memory: 50M
```

Autoscaling

What's better than running a command to scale up a service? Running *no* commands to scale up a service. This is known as autoscaling. It involves monitoring key usage and load metrics for containers and detecting when they get close to becoming overloaded. At this point, the new containers for the service are launched to meet with additional demand. As the load dies down, this is again detected, and the service is scaled back down, freeing up more resources.

Unfortunately, Swarm does not provide built-in autoscaling, although it's possible to engineer it yourself with each node in the cluster exporting metrics (using something like cadvisor[3] to a central metric service, such as Prometheus[4]). If that sounds like too much work, you could consider one of various open source solutions, such as Orbiter.[5] Your final option is to switch to using Kubernetes[6] for your container orchestration. Although more complex, it is more fully featured and has autoscaling built in.[7]

Zero-Downtime, Blue-Green Deploys

The pinnacle of a good continuous deployment pipeline is being able to deploy updates to your app in a seamless, safe way, with no downtime or impact on users. Typically, this is achieved with *blue-green* deploys, where a second version of the application is started, and then traffic is (usually gradually) cut over to the new version of the app. The previous version of the app is kept around (at least for a while) in case a problem emerges that means you need to roll back.

Docker Swarm provides some capabilities for performing these type of rolling updates.[8] While this can be useful, unfortunately Swarm currently doesn't

3. https://github.com/google/cadvisor
4. https://prometheus.io
5. https://github.com/gianarb/orbiter
6. https://kubernetes.io
7. https://kubernetes.io/docs/tasks/run-application/horizontal-pod-autoscale/
8. https://docs.docker.com/engine/swarm/swarm-tutorial/rolling-update/